Auto Power Switching PCBA Specification

Purpose

Automatically switch between two 12V DC power sources:

- Input 1 (AC Mains): 300W AC-DC PSU (12V @ 25A max)
- Input 2 (DC Source): 12V–48V input via DC-DC PSU (also 12V @ 25A max output)

The switching board should route only one 12V source at a time to the main power block based on which is active or preferred.

Electrical Specifications

- Input Voltage (both sources): 12V DC (±5%)
- Max Current per Channel: 25A
- Max Total Load: 300W
- Switching Mode:
    - Automatic seamless failover
    - Prefer AC-derived source (if both are present)
- Protection:
    - Backfeed protection (no reverse current into unused PSU)
    - Overcurrent / thermal shutdown preferred
- Switching Delay: <10 ms preferred
- Efficiency: >99% (pass-through type preferred)

Connectors / Interface

- Input 1 (AC PSU 12V Out): Screw terminal (12V IN1, GND)
- Input 2 (DC PSU 12V Out): Screw terminal (12V IN2, GND)

- Output to System Power Block: Screw terminal (12V OUT, GND)

Preferred Architecture

- MOSFET-based power path controller with ideal diode functionality (e.g., using something like TI's LM5050, TPS2121, or similar)
- Minimal voltage drop across the FETs
- Heat-sinking or thermal pad mounting option recommended